The impact of the minimum price per unit of alcohol in Scotland for alcohol consumption

To what extent has the minimum price per unit in Scotland been effective in reducing consumption and demand for alcohol since 2018?

Economics

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Introduction

On May 1 2018 the Scottish government implemented a new microeconomic strategy in the hopes of reducing the consumption of alcohol by setting a minimum price per unit (MUP) of alcohol at 50p per unit, making it the lowest price an alcoholic drink can be sold for (The Lancet, 2021). A unit of alcohol is equal to 10 ml or 8 grams of pure alcohol (NHS, 2021). Prior to this change, the lower-risk weekly drinking guidelines amount of 14 units could be bought for as low as approximately £2.50 but with the MUP, it can't be sold lower than £7 (Scottish Government, 2018). The MUP applies to all types of alcoholic drinks such as spirits, wines, beer and cider, but mainly targets cheap off-trade drinks (i.e. drinks sold elsewhere than in bars, restaurants and clubs) containing higher concentrations of alcohol which encourages excessive drinking while having no effect on drinks already above the minimum amount (Alcohol Focus Scotland, 2018). This is particularly important as alcohol is a negative externality of consumption when over consumed, and it's become an increasingly prominent issue as alcohol became 64% more affordable since 1980 (Alcohol Focus Scotland, 2018). This resulted in more people drinking potentially leading to more alcohol related health and social problems. Data already demonstrates alcohol-related mortality rates are more than twice as high as they were in the 1980s, despite major improvements in health care (Alcohol Focus Scotland, 2018). Additionally, alcohol related health issues use up a hospital's valuable resources and time imposing a further cost on society. Consumption of alcohol is a sensitive social and economic issue as it's associated with social and convivial occasions but is both addictive and difficult to substitute. The relationship between price and consumption for alcohol will be further explored in this essay as well as the effectiveness of the MUP since its introduction.

Methodology

This essay investigates the impact of the MUP on the demand and consumption of different types of alcohol, over the time it has been introduced in Scotland. There are many different key aspects to consider in order to properly evaluate the effectiveness of this alcohol policy. As Scotland is the first to introduce an MUP, this policy is heavily scrutinised by many different sources. Government sources were easily accessible to determine the social impacts and changes in consumption in the off-trade, attributed to the MUP. Other sources were used to determine the price elasticity of demand (PED), the average change in price for different drink types, and the market share for each type of alcoholic beverage. Furthemore, economic concepts such as the price mechanism and the law of demand were applied to analyse and evaluate these statistics. This essay will lso investigate why an MUP was chosen as the intervention tool as it's the first of its kind. Finally, economic concepts such as inflation, cross elasticity of demand (XED), and behavioural economics, were applied to analyse the limitations behind the implementation of the MUP.

Evidence 1: The effect of the MUP on alcohol demand and consumption

The table below summarises some academic research that has been done to calculate the PED for the various categories of alcoholic drink. The studies yield remarkably similar results (in particular for wines, spirits and alcohol) and in each case, the PED of each type of drink stated in the table is inelastic as the values are all below the absolute value of 1. What this means is a change in price leads to a proportionately smaller change in quantity demanded. For example, using the Gallet 2007 PED value for beer, a 1% increase in price only leads to a 0.36% decrease in quantity demanded.

Table 1: Price elasticity of demand of alcoholic drinks (PED) (Robson, 2012)

Type of drink	PED Values
Wine	-0.700 (Gallet, 2007) -0.68 (Clements et al, 1997) -0.69 (Wagenaar, 2009)
Spirits	-0.679 (Gallet, 2007) -0.98 (Clements et al, 1997) -0.8 (Wagenaar, 2009)
Beer	-0.36 (Gallet, 2007) -0.35 (Clements et al, 1997) -0.46 (Wagenaar, 2009)
Alcohol	-0.497 (median: Gallet, 2007) -0.51 (Wagenaar et al, 2009)

The inelasticity of each drink is due to a few reasons:

1. There are no close substitutes to any alcoholic drink, making it a unique product;
2. "Personal effect motives", typically defined as the tendency to use alcohol as a means to escape, avoid, or regulate unpleasant emotions (NCBI, 2015);
3. Bounded self-control of human beings due to the addictive nature of alcohol;
4. Utility maximisation from drinking alcohol due to its effects;

5. Compared to the other types of alcohol mentioned in the table, beer is more inelastic due to it typically being seen as a common beverage consumed during every day leisure activities (euro.who, n.d.)

One of the critical theories of economics is the law of demand, which states as prices increase the quantity demanded will fall and vice versa. The price inelasticity due to the factors stated above may question the overall effectiveness of the MUP as there are concerns increasing the price won't influence consumer behaviour to the extent the government is hoping it will. If this were to be the case, it would solely make alcohol less affordable for those within the lowest income brackets, which will only lead to further problems as more money will be spent on the same quantity of alcohol. Regardless, increasing the price of alcoholic beverages has been deemed one of the most effective interventions to minimise harmful use of alcohol intake, by the World Health Organisation in 2019 (WHO, 2019).

By examining alcohol by volume (ABV) for each type of alcoholic drink it can be roughly determined how they'll get impacted by the MUP. Beer has the lowest ABV, containing about 4-6%, followed by wine at less than 14%, and lastly spirits at 37% (alcohol rehab guide, 2022). With these numbers coupled with the PED values from the table it's safe to assume:

1. Spirits will see the biggest change in quantity demanded and price as they have the highest ABV and elasticity;
2. Wine has lower ABV so there will be less units of alcohol in the drink, meaning the price will increase less significantly
3. Beer should see the smallest changes in both price and consumption out of all alcoholic beverages as it has low ABV and is most inelastic.

The above elasticities align well with the hopes of the Scottish government to reduce consumption of alcohol for harmful and hazardous drinkers, as it was reasoned the most

harmful drinks (ie highest ABV) are those that would see the biggest reduction due to the MUP.

Table 2: Price changes of different alcoholic beverages in pence per gram (BMJ, 2019)

Drink types	Price (pence)	Grams bought per adult per household by week
Beer		
Change	0.594	-2.92
Wine		
Change	0.420	-1.67
Spirits		
Change	0.670	-2.89
Fortified Wine		
Change	0.911	0.32
Cider		
Change	1.448	-2.34

The table above conveys the immediate impact of the MUP in Scotland on the average price per gram in pence and on the average grams bought per adult per household by week. From this, it's possible to determine exactly how prices have been affected by the MUP for different types of drinks. Changes in the price were broadly similar across beers (0.594), wines (0.420), and spirits (0.670). Out of these three, spirits were understandably the highest, and, surprisingly, beers saw a larger change in price than wine did. This may be as the data includes both off-trade and on-trade prices. On the other hand, cider and fortified wines saw much larger increases in price compared to the others.

Based on the statistics mentioned above, by applying the law of demand, it can be confidently presumed each alcoholic beverage mentioned will decrease in quantity

demanded and, likewise, consumption should decrease as well. This is due to the income effect, which occurs when the price of a good increases, resulting in a decrease of quantity demanded as consumers are spending a greater proportion of their income, making it less affordable.

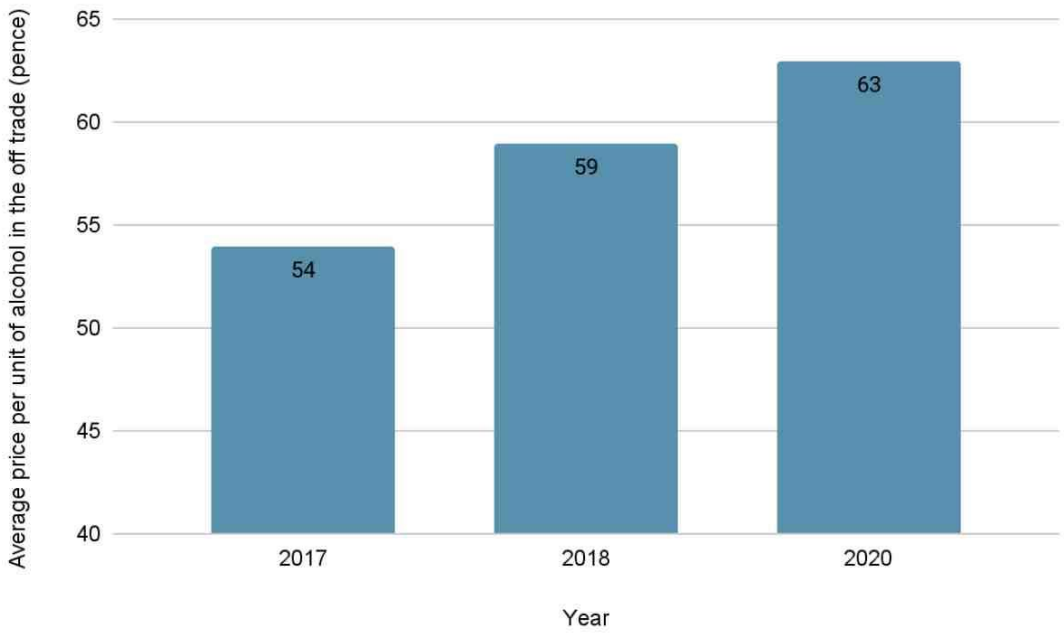
In order to accurately establish the relationship between price and quantity consumed for alcohol, it's necessary to observe the statistics below before the implementation of the MUP and then compare it to what is found after. Scotland's Monitoring and Evaluating Scotland's Alcohol Strategy (MESAS) presents the most recent available data on alcohol in Scotland as well as information from Public Health Scotland. By using these sources across 3 different years, in 2017 (1 year before the MUP), another in 2018 (immediately after the MUP), and finally 2020 (medium term), we can accurately acquire the information needed to determine the effectiveness of the MUP for the immediate and longer-term impact. A few key data points from each year, will be extracted from the reports, including:

1. Average price per unit of all alcohol in the off trade
2. Total litres of pure alcohol sold in Scotland and Litres of pure alcohol sold per adult
3. The market share of different types of alcoholic beverages and the effect the MUP has had on them

Beginning with a bar graph to easily compare and portray the average price per unit of all alcohol in the off trade.

Figure 1: Average price per unit of alcohol sold

Data source: MESAS reports 2017-2020 (bar graph is author made)



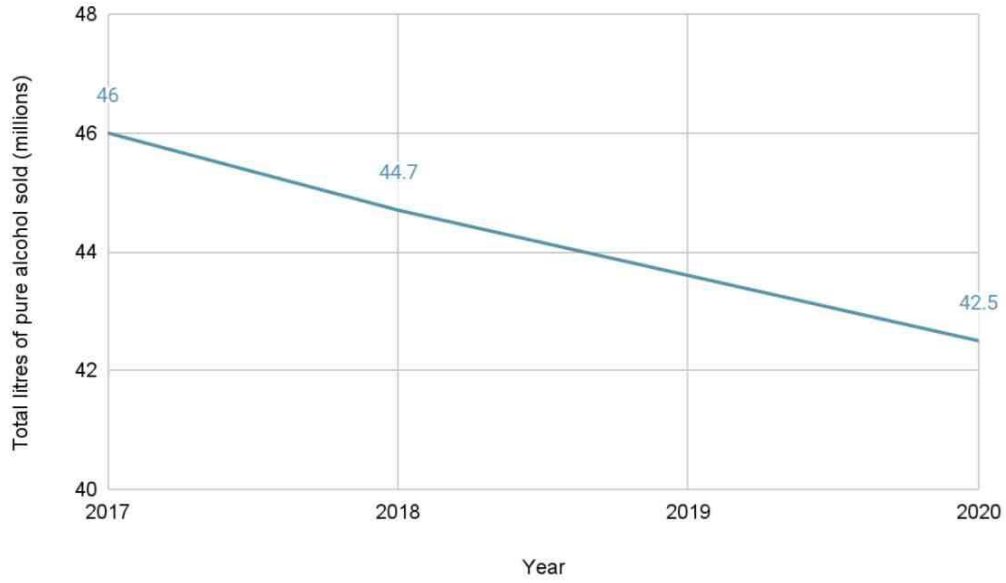
The bar graph illustrates continuous increases of the average price per unit of alcohol throughout all three years. This is a crucial statistic, because a price is what drives consumption. So by making alcohol less affordable, consumption is expected to decrease, despite alcohol's relative price inelasticity. In 2017 the average price per unit of all pure alcohol sold in off trade specifically, was estimated to be about 54 pence. Just under half of all off trade alcohol (47%) was sold at below what the minimum price would become a year later, at 50 pence per unit. Furthermore, during the year before (2016), the corresponding numbers were 52 pence and 51% respectively. These numbers alone represent the biggest yearly changes in average off-trade prices since 2013. Which makes it more impressive the following year in 2018, the average price per unit increased significantly to 59 pence and 23% of all pure alcohol sold was below 50 pence (please note that although the MUP was introduced the same year, there were still 4 full months before its introduction, which explains why nearly a quarter of alcohol was sold below 50 pence). This demonstrates the significant immediate impact of the MUP in only its first 8 months since implementation by showing some of the biggest changes in alcohol off-trade prices in years. In addition, two years after the implementation of the MUP, alcohol prices continue to show increases, as the

average price per unit was 63p. 64% of the total pure alcohol sold during this year, was reported as being between 50.0p and 64.9p per unit, a very large increase from the 32% it was before the introduction of the MUP.

Moving forward, a line graph was chosen in order to clearly portray a trend over the three years for total litres of pure alcohol sold in Scotland

Figure 2: total litres of pure alcohol sold in all of Scotland

Data source: MESAS report 2017-2020 (line graph is author made)



In 2017 there was a reported 46 million litres of pure alcohol sold in Scotland. This value is especially important to consider in order to set the foundation for comparison when reviewing the other data sets, as well as to accurately determine whether or not the MUP has had a significant effect if any. In the same year, 10.2 litres of alcohol were sold per adult, equal to 19.6 units of alcohol per adult per week. This is a similar value when compared to what it was in 1994. In contrast, in Wales and England, per adult sales in Scotland were 14% higher, because more alcohol was sold in the off trade much cheaper in Scotland. The following year, along with the implementation of the MUP, total litres of alcohol sold were at

44.7 million, the same as 19 units of alcohol per adult per week. These alcohol sales in 2018 were reported to be at their lowest point in the last 25 years (Alcohol Focus Scotland, 2020), establishing a beneficial and significant impact of the MUP, even without a full year passing by. As further proof of its beneficial impact, 9% more alcohol was sold per adult in Scotland compared to Wales and England, a significant decrease from 14% from the previous year. Finally, in 2020, the downward trend continues as seen on the line graph; the total litres of alcohol sold during the year was at 42.5 million, equal to 18 units per adult per week. This sets another all-time low in Scotland for alcohol consumption, since 1994. Moreover, the volume of alcohol sold per adult in Scotland was 6% higher than in England and Wales, the smallest difference between these regions since 1994.

By combining information from figure 1 and 2 it's possible to directly calculate the PED of alcohol in Scotland:

$$PED = \frac{\%\,\Delta\ in\ Qd\ of\ Alcohol\ in\ Scotland}{\%\,\Delta\ in\ P\ per\ unit\ of\ alcohol\ in\ Scotland}$$

PED calculations:

$$\%\,\Delta\ in\ Qd = \frac{46 - 42.5}{46} \times 100$$

$$= 7.6087\%$$

$$\%\,\Delta\ in\ P = \frac{54 - 63}{54} \times 100$$

$$= -16.6667\%$$

$$PED = \frac{7.6087\%}{-16.6667\%}$$

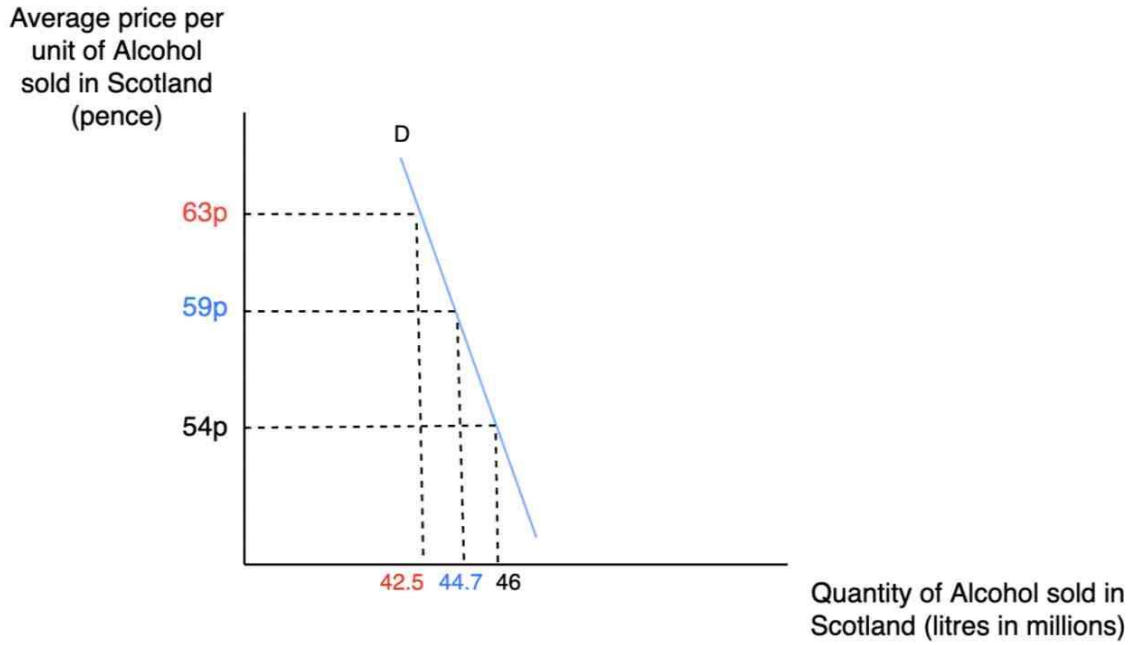
$$PED = -0.46\ (2\ d.p)$$

The PED value states a 1% increase in price leads to an approximately 0.46% decrease in quantity demanded, a relatively inelastic value. This provides a generalised PED value as this groups all alcohol types together and it's expected each type of alcohol would vary in

elasticity. When comparing this to the alcohol PED in table 1, the values are fairly similar despite the 10+ year time difference. However, the calculation cannot assume ceteris paribus as it's unsure how much the change in Qd can be attributed to the MUP. For example, alcohol campaigns and negative advertising may have impacted the decrease in Qd.

Furthermore it's also possible to create a demand curve by combining the data from figure 1 and 2. Figure 3 below visualises the effects of the alcohol market in Scotland as a result of the MUP.

Figure 3: Demand curve of alcohol in Scotland (author made)



The demand curve has a steep gradient due to its price inelasticity. The increase in price leads to a proportionately smaller decrease in demand emphasised by the space between the prices is much larger than the space between the quantity of alcohol sold. Nonetheless, the increase in price led to decreases in demand as evidence of the effectiveness of the policy.

Additionally, the MUP has shifted the market for different types of alcoholic beverages. It's possible to assume ciders and spirits should see a large decrease in market share, as they're

both more elastic and saw more significant changes in price. Whereas beer and wine may see a rise in consumption as they're the substitutes to the stronger, more expensive alcoholic beverages and are more inelastic

Table 3: The MUP's effect on drinks' market share

(Public Health Scotland, 2021)

Drink type	Market share (%)	MUP effect (%)
Spirits	32.5	-4.7
Wine	31.9	1.3
Beer	24.0	-1.9
Cider	6.4	-18.8

Based on the information from the table, spirits and particularly cider experienced a much larger decrease in market share at -4.2% and -18.8% respectively. Wine saw less significant positive changes at 1.3%, which can be justified by the fact that it's the next most alcoholic drink after spirits and saw slightly lower price changes as seen in table 2, so naturally people will switch to wine instead. Although surprisingly, beer experienced a decrease in market share despite it being the most inelastic beverage out of all of the others, at an average price elasticity of demand of 0.39. Generally, these are the type of results the Scottish government would have wanted to see in order to determine whether or not the MUP is a valuable intervention tool, as hoped for, consumption of heavily alcohol concentrated drinks have decreased.

Evidence 2: The effectiveness of a minimum price as a solution

Scotland has had persistent problems with alcohol consumption. For every adult in Scotland, 11.8L of pure alcohol are sold each year, compared to England's smaller 9.6L, therefore meaning Scotland's consumption of alcohol is 23% higher than in England (Robson, 2012). Additionally, the negative health related impacts of alcohol over the last 20 years have increased in Scotland such as chronic liver disease and cirrhosis, whereas in most parts of Western Europe these impacts are reported to have fallen (NIHR, 2021) (Robson, 2012). Furthermore, in the off-trade in particular, alcohol has become much more affordable since 1987 as a result of rises in income and inflation. In an attempt to reduce alcohol related harms and the consumption of alcohol, Scotland was the first country in the world to implement a minimum unit price for alcohol as opposed to other solutions like increasing VAT and excise duties, which most other countries targeting alcohol have done.

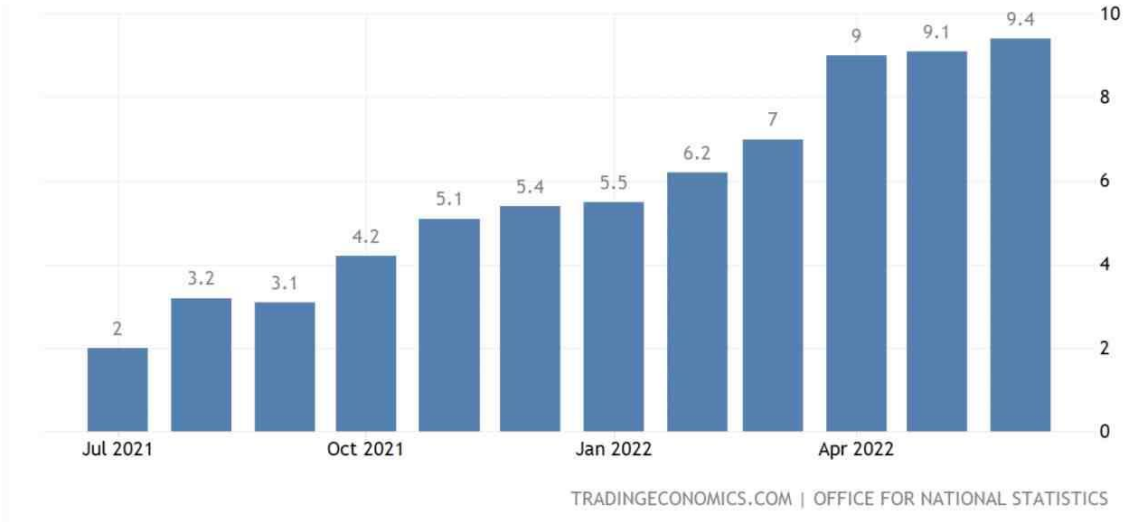
The majority of alcohol's negative externality comes from excessive drinking and not those drinking more casually. Whereas a tax would affect the entire market evenly, a minimum price only affects drinks sold below the price floor, therefore, there is no effect on more expensive drinks. For this reason a minimum price is particularly effective at reducing consumption of alcohol drunk by heavy drinkers, as they tend to consume cheap alcohol with high amounts of units (Scot Gov, 2018). In addition, the minimum price is unlikely to affect on-trade sales at bars and restaurants, as the alcoholic drinks typically sold in those places are already above the minimum price. In fact, in Scotland the average price per unit in the on-trade was £1.80 (Alcohol Focus Scotland, 2020). This further reinforces the justification for a minimum price as most alcohol is consumed excessively off-trade because supermarket alcohol can be sold cheaply. Once again this neither affects those that wish to drink casually nor does it undermine the social aspect of drinking at a pub or restaurant. Finally, the minimum price per unit of alcohol is also attractive as it impacts the younger drinkers as well (NCBI, n.d). For the same reason the MUP affects the heavy drinkers, it also targets younger

drinkers as they typically buy the cheapest alcohol off trade. This is especially important as young drinkers are more vulnerable to the risks of drinking large quantities of alcohol, compared to adults. As the brain continues to develop up until ages 20-25, drinking at younger ages may significantly increase the risk of damage to the developing brain and may potentially lead to more alcohol related problems later in life (Better Health, n.d.).

Evidence 3: What limits its effectiveness

The rising inflation in the UK, which as of April 2022 was 9.4% (rateinflation, 2022), will undermine the impact and severity of the 50p minimum price since its introduction in 2018 (SHAAP, 2021). The graph below shows the ever-increasing trend for the inflation rate in the UK since July 2021.

Figure 4: Inflation rate in the UK 2021-2022
Source: Trading Economics, 2022



Inflation is defined as the persistent increase in the average price level in the economy. Because the price levels in the economy continue to rise at such a high level, but the minimum price is a set amount, the value of the MUP starts to get significantly eroded. Therefore, the MUP should be rising in line with inflation to maintain its relative effectiveness. In response to this, 28 organisations are calling to raise the MUP without delay, to a more desirable 65p (BBC, 2021).

In some cases the MUP has had little impact on decreasing the demand for alcohol among low income and the heaviest of drinkers. Based on a series of qualitative interviews from the Chief Scientist Office (CSO), there are varying reasons as to why the MUP has failed to influence these consumers' behaviour in particular, such as:

1. Some people switched to other alcoholic drinks, most notably from strong white cider to spirits such as vodka, meaning the MUP may have incentivised some positive changes in behaviour, such as what and how some people drink, but not the quantity they would consume
2. The MUP does not tackle the core issues behind the reasons as to why people drink in excess e.g. trauma or the hardships of life
3. Inadequate provision of services to do with mental health.
4. Bounded self control due to the addictive nature of alcohol

Each of these together threaten the overall efficacy of what the Scottish Government had hoped to achieve, especially as all the factors mentioned above have to do with the demographic the government had hoped to address. With external factors such as the natural bounded self control of human beings and a substitution effect to other alcoholic beverages, this proves there is more to decreasing consumption and demand for alcohol than just raising prices indicating the government may have to widen their scope if they wish to see more significant results.

In addition, although the impact from this may be relatively small, it's worth mentioning there has been evidence of people purchasing alcohol by crossing borders to places (e.g. England) where they can buy alcohol without an MUP in place. This would apply primarily to individuals living closer to the border and can take advantage of trips to do bulk purchasing. To serve as reference, the estimated population near the borders of Scotland is 115,270, an estimated 2.11% of Scotland's total population of 5.454 million (ONS, 2018). Bulk purchasing is done in order to save as much money as possible without the MUP and to offset the costs of transportation or travel. In the off-trade 12 months after the implementation of the MUP, in the North East and North West of England, closest to the border of Scotland, there was an increase in sales of 1.14% (Public Health Scotland, 2022). This suggests the effect of cross elasticity of demand (XED), the measurement of the responsiveness of quantity demanded

for one good to the change in price of another good, which in this context would be the increase in quantity demanded of alcoholic beverages in the UK due to an increase in price of alcoholic drinks in Scotland. Additionally, 3% of adults living in the area have admitted to crossing borders in order to take advantage of the cheaper alcohol (Public Health Scotland, 2022). Once again, the extent of cross border purchasing appears to be currently small, regardless, it still does somewhat undermine the effectiveness of the policy.

Conclusion

The data in Scotland pointed to increasingly worrying alcohol-related negative externalities of consumption spurring the Scottish government to evaluate policies to combat the trend. Despite alcohol being relatively price inelastic, increasing the price of an alcoholic beverage has been deemed an effective tool to influence and minimise harmful use of alcohol intake and the Scottish government attempted to use price in a targeted way via the introduction of the MUP. Since the implementation of the MUP in 2018, a downward trend in the consumption of alcohol as a result of higher prices reveals a positive outcome for the Scottish government. This strongly suggests the effectiveness of the MUP as some of the largest changes in decades in price changes and total consumption of alcohol can be directly linked to the MUP. There are also compelling arguments supporting the rationale behind choosing an MUP over a different type of intervention such as taxes and excise duties. The tool has considerable advantages in terms of ease of introduction and enforcement and a minimum price is efficient when it comes to targeting a specific demographic, which in this case would be harmful and hazardous drinkers. It enables the government to target the cheaper and more harmful drinks in terms of alcohol concentration, impacting the people who abuse those drinks the most while not significantly affecting those drinking casually. However, if the government wishes to maintain and enhance this positive impact, further steps are required such as raising the MUP to counter inflation, tackling the causes of over-consumption of alcohol and directly helping those with alcohol dependency with the necessary services

Evaluation

Despite the high reliability of the sources used to conduct this investigation, all information was limited to what could be found online resulting in certain issues. One issue is that there are variations in methodology from source to source, making it difficult to decide what source would be more accurate. This is the reason why when investigating a certain variable such as PED or changes in consumption the same source was used in order to ensure more consistency. Fortunately, the MUP is one of the most in-depth investigated policies in Scotland's history. This led to a wide array of data sets from not only external sources, but many government sources as well. Additionally, it should be noted the scale of this investigation is rather large as it attempts to determine the effect of a single policy on an entire market. With the large variety of alcoholic drinks, coupled with the fact that they each have their own price elasticities of demand, means each drink reacts differently to the MUP and, therefore, some may not influence changes in behaviour. Despite this, the effects on each alcoholic drink were generalised in parts of this investigation. Naturally, there could also be many other separate reasons for the decrease in consumption of alcohol, over the last few years since the implementation of the MUP. Things such as increasing public awareness of the dangers of excessive drinking like the 'Choices for Life' campaign as well as the many others like the 'It'll cost you!" and 'Don't Buy it, Don't Supply it' campaigns, which all strive for broadening awareness and reducing alcohol consumption in Scotland (Choices For Life, n.d.) (Police Scotland, 2022) (Scottish Borders Council, 2022).

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